

Lecture Notes on Numerical Methods

Error & Convergence

IIITDM Kancheepuram, Chennai, India

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Evaluation (Tentative)

Mid-semester: 25

Project: $25 = 10$ (*probleme presentation*) + 15 (*final deliverables*)

End - semester : 50

Textbooks and References

Textbooks:

1. David Kincaid, Ward Cheney; Numerical Analysis: Mathematics of Scientific Computing, Third Edition. American Mathematical Society, 2012. (ISBN: 978-0-8218-4788-6).
2. Brian Bradie; A Friendly Introduction to Numerical Analysis; Pearson Education India. 2007.(ISBN: 9788131709429)

References:

1. Kendall E. Atkinson. "An Introduction to Numerical Analysis." John Wiley & Sons, Inc., 1988.
2. Curtis F. Gerald, Patrick O. Wheatley; "Applied Numerical Analysis", Pearson Education India; 7th edition (2007).

Syllabus

Error and Convergence: Type of error, Computer representation of numbers, rate of convergence, and order of convergence.

Root findings: Bisection, Newtons, Secant Method, Newtons Method, Fixed-point Iterations, Methods for multiple root case.

Interpolations: Lagrange, Newton, Divided Difference, Chebyshev polynomials. Piecewise interpolations and extrapolations.

Linear Equation and Eigenvalue: Direct and Iterative methods, Norms, Orthogonality, QR Decomposition, Power Method.

Difference and Differential Equations: Difference formula, Methods for ODE, Stability, and error analysis.

Integrations: Trapezoidal, Simpson and Gauss Quadrature, .

Finite Difference Method: Two point BVPs.

Floating Point Approximation

Definition (n-digit Floating Point)

Let $\beta \in \mathbb{N}$ & $\beta \geq 2$, an n-digit floating point number in base β is of the form:

$$\pm q \times \beta^e = (-1)^s \times (\cdot d_1 d_2 \cdots d_n)_\beta \times \beta^e$$

where

$$(\cdot d_1 d_2 \cdots d_n)_\beta = \frac{d_1}{\beta} + \frac{d_2}{\beta^2} + \cdots + \frac{d_n}{\beta^n}$$

where $d_i \in \{0, 1, \cdots, \beta - 1\}$ with $d_1 \neq 0$ and q is mantissa.
 $s = 0$ or 1 and appropriate exponent e . β is called the radix.

► $\beta = 2$ is Binary

► $\beta = 10$ is Decimal

Note: *Computer works internally in the Binary system but communicates with its human users in the Decimal system.*

Example

1. $x = 6.238 = (-1)^0 \times 0.6238 \times 10^1$
here $s = 0, \beta = 10, e = 1, d_1 = 6, d_2 = 2, d_3 = 3, d_4 = 8$ is decimal floating point representation.
2. $x = -0.0014 = (-1)^1 \times 0.14 \times 10^{-2}$ is represented in the decimal floating point where
 $s = 1, \beta = 10, e = -2, d_1 = 1$ and $d_2 = 4$.

Remark:- Floating point representation of $\frac{1}{3}$
 $\frac{1}{3} = 0.333\cdots = (-1)^0 \times (0.333\cdots)_{10} \times 10^0$

Limitations:-

- ▶ Number of digit in Mantissa (n)
- ▶ Range of Exponent (e)

Underflow and Overflow of Memory

$$m \leq e \leq M$$

where e is an exponent.

- ▶ $e > M \rightarrow$ Memory Overflow $\rightarrow$ inf and NaN (Not a Number)
- ▶ $e < m \rightarrow$ Memory Underflow $\rightarrow (0 \times \infty, \infty - \infty, 0/0, \dots)$

Output NaN

X is sufficiently large (Overflow)

x is sufficiently small (Underflow)

$$X \cdot x = \text{NaN (Not a Number)}$$

- ▶ Single Precision: $\beta = 2, n = 24$

$$\text{e-range} = \begin{cases} m = -126 \\ M = 127 \end{cases}$$

► Double Precision: $\beta = 2$, $n = 53$

$$\text{e-range} = \begin{cases} m = -1022 \\ M = 1023 \end{cases}$$

$m = -\infty$ and $M = \infty$ for the representation of any conceivable real number.

m & M is finite for Finite Precision numbers

Exercise:-

Show that Number of floating - point numbers for a normalized system with parameters (β, n, m, M) is

$$N = 2(\beta - 1)\beta^{n-1}(M - m + 1) + 1$$

Definition of Normalized Floating - point System

The system is said to be normalized when $d_1 = 0$ if and only if $x \neq 0$

Rounding of a Decimal Number

x is a positive decimal number of the form $0.\square\square\cdots\square$ with m digits to the right of the decimal point.

One rounds x to the n -decimal ($n < m$) in a manner that depends on the value of the $(n+1)^{\text{th}}$ digit.

- ▶ If $(n+1)^{\text{th}}$ digit is 0, 1, 2, 3 or 4 the n^{th} digit is not changed.
- ▶ If it is 5, 6, 7, 8 or 9 then the n^{th} digit is increased by 1- unit and remaining digits are discarded.

Note 1

x is rounded $\rightarrow \tilde{x}$, which is an n -digit approximation, then

$$|x - \tilde{x}| < \frac{1}{2} \times 10^{-n}. \quad (1)$$

Proof

Case I: If $(n + 1)^{\text{th}}$ number of x is 0, 1, 2, 3 or 4, then

$$x = \tilde{x} + \epsilon$$

$$|x - \tilde{x}| < \epsilon \times 10^{-n}.$$

So, Eq (1) follows.

Case II: If $(n + 1)^{\text{th}}$ number of x is 5, 6, 7, 8 or 9 then

$$\tilde{x} = \hat{x} + 10^{-n}$$

Where $\hat{x}$ is a number with same n -decimal digits as x and all digits beyond the n^{th} are 0.

$$x = \hat{x} + \delta \times 10^{-n} \quad \text{with} \quad \delta \geq \frac{1}{2}.$$

$$\tilde{x} - x = (1 - \delta) \times 10^{-n}$$

Proof contd.

Again

$$\delta - 1 \geq -1 + \frac{1}{2} \Rightarrow 1 - \delta \leq \frac{1}{2}$$

Hence

$$\tilde{x} - x \leq \frac{1}{2} \times 10^{-9} \quad (2)$$

So, Eq (2) follows Eq (1)



Note 2

Chopped or Truncated n - digit approximation

$$|x - \hat{x}| < 10^{-n}$$

Prove this.

Error

Error = True Value - Approximated Value

If x is true value and $\tilde{x}$ is approximated value then Error,

$$E(x) = x - \tilde{x}$$

Types of Errors

1. Absolute Error: $E_a(x) = |x - \tilde{x}| = |E(x)|$
2. Relative Error: $E_r(x) = \frac{E(x)}{x}$ for $x \neq 0$
3. Percentage of Error : $E_P(X) = 100 \times |E_r(x)|$

Example 1

let $x = 100000$ and $\tilde{x} = 99999$

then $E_a(x) = 1$, $E_r(x) = 10^{-5}$, $E_P(x) = 10^{-3} \%$

Example 2

let $x = 1$ and $\tilde{x} = \frac{1}{2}$

then $E_a(x) = \frac{1}{2}$, $E_r(x) = \frac{1}{2}$, $E_P(x) = 50 \%$

Error contd.

Theorem (Roundoff error on adding)

Let $x_0, x_1, \dots, x_n$ be positive machine number and each have unit roundoff error is ϵ , then relative error is computed by

$$\sum_{i=0}^n (x_i) \text{ is usual way is at most } (1 + \epsilon)^n - 1 \approx n\epsilon$$

Proof:

Let $S_k = x_0 + x_1 + \dots + x_k$ and S_k^* be what computer calculated instead of S_k .

The recursive formula

$$S_0 = x_0 \text{ and } S_{k+1} = S_k + x_{k+1}$$

$$S_0^* = (x_0) \text{ and } S_{k+1}^* = \text{fl}(S_k^* + x_{k+1})$$

Proof contd.

For the analysis we define ρ_k and δ_k

$$\rho_k = \frac{S_k^* - S_k}{S_k} \text{ and } \delta_k = \frac{S_{k+1}^* - (S_k^* + x_{k+1})}{S_k^* + x_{k+1}}$$

Thus $|\rho_k|$ is the relative error in approximating the k^{th} partial sum S_k^* and

$|\delta_k|$ is relative error in approximating $S_k^* + x_{k+1}$ by quantity $\text{fl}(S_k^* + x_{k+1})$

$$\begin{aligned}\rho_{k+1} &= \frac{S_{k+1}^* - S_{k+1}}{S_{k+1}} \\ &= \frac{(S_k^* + x_{k+1})(1 + \delta_k) - S_{k+1}}{S_{k+1}}\end{aligned}$$

Proof contd.

$$\begin{aligned}\rho_{k+1} &= \frac{(S_k(1 + \rho_k) + x_{k+1})(1 + \delta_k) - (S_k + x_{k+1})}{S_{k+1}} \\&= \frac{(S_k + S_k\rho_k + x_{k+1})(1 + \delta_k) - (S_k + x_{k+1})}{S_{k+1}} \\&= \frac{\cancel{S_k} + S_k\rho_k + \cancel{x_{k+1}} + S_k\delta_k + S_k\rho_k\delta_k + x_{k+1}\delta_k - \cancel{S_k} - \cancel{x_{k+1}}}{S_{k+1}} \\&= \frac{\delta_k(S_k + x_{k+1})}{S_{k+1}} + \frac{\rho_k S_k(1 + \delta_k)}{S_{k+1}} \\&= \delta_k + \rho_k \left(\frac{S_k}{S_{k+1}} \right) (1 + \delta_k) \quad \text{Here } S_k + x_{k+1} = S_k + 1\end{aligned}$$

Since $S_k < S_{k+1}$ and $|S_k| \leq \epsilon$

$$|\rho_{k+1}| \leq \epsilon + \rho_k(1 + \epsilon)$$

$$= \epsilon + \theta(\rho_k) \text{ and set } \theta = 1 + \epsilon$$

Proof contd.

$$|\rho_{k+1}| \leq \epsilon + \theta(\rho_k)$$

Put $k = 0, 1, 2, \dots$

We have the following successive inequalities:

$$|\rho_0| = 0$$

$$|\rho_1| \leq \epsilon$$

$$|\rho_2| \leq \epsilon + \theta\epsilon$$

$$|\rho_3| \leq \epsilon + \theta(\epsilon + \theta\epsilon)$$

$$\vdots$$

The general result is

$$\begin{aligned} |\rho_n| &\leq \epsilon + \theta\epsilon + \theta^2\epsilon + \dots + \theta^{n-1}\epsilon \\ &= \epsilon(1 + \theta + \theta^2 + \dots + \theta^{n-1}). \end{aligned}$$

Proof contd.

$$|\rho_n| \leq \epsilon \left(\frac{\theta^n - 1}{\theta - 1} \right) \text{ put } \theta = 1 + \epsilon$$

$$= \epsilon \frac{(1 + \epsilon)^n - 1}{\epsilon}$$

$$= (1 + \epsilon)^n - 1$$

$$= 1 + \binom{n}{1}\epsilon + \binom{n}{2}\epsilon^2 + \cdots + \epsilon^n - 1 \text{ (by the binomial theorem)}$$

$$= n\epsilon + o(\epsilon^2)$$

$$\approx n\epsilon$$



Convergence

Definition

The sequence $\{x^n\}$ converges to the value L provided

$$\lim_{n \rightarrow \infty} x^n = L$$

or, equivalently

$$\lim_{n \rightarrow \infty} |x^n - L| = 0$$

Type of Measure of Convergence: There are two measure of convergence. (Related to Speed)

1. **Order of convergence**
2. **Rate of Convergence**

Order Of Convergence

Example

$$f(x) = \frac{\cos x - 1 + \frac{1}{2}x^2}{x^4}$$

What is the limit of $f(x)$ as $x \rightarrow 0$?

Solution:

$$f(x) = \frac{\cos x - 1 + \frac{1}{2}x^2}{x^4} = \frac{1}{24} - \frac{1}{720}x^2 \cos \zeta$$

for some ζ between 0 and x .

Order Of Convergence

Solution Contd.

$$\left| \frac{\cos x - 1 + \frac{1}{2}x^2}{x^4} - \frac{1}{24} \right| = \frac{1}{720} |x^2 \cos \zeta| \leq \frac{1}{720} |x^2|$$

it follows that $\lim_{n \rightarrow \infty} f(x) = \frac{1}{24}$ and the rate of convergence is $O(x^2)$.



Note: Let $\{p_n\} \rightarrow p$ and $e_n = p_n - p$ then Rate of convergence deal with p_n and p while Order of convergence deal with successive error value

Big O Notation

- ▶ Big O notation is a mathematical concept used to describe the asymptotic behavior of functions.
- ▶ It classifies functions according to their growth rates and is widely used in calculus, numerical analysis, and computer science.

Definition: A function $f(x)$ is said to be $O(g(x))$ (read as “big O ” of $g(x)$) if there exist constants $C > 0$ and x_0 such that:

$$|f(x)| \leq C \cdot |g(x)| \quad \text{for all } x \geq x_0.$$

- ▶ **Algorithm Analysis:** Big O is used to describe the time or space complexity of algorithms.
 - ▶ Example: A sorting algorithm with time complexity $O(n^2)$ means the runtime grows quadratically with input size n .
- ▶ **Approximation:** In numerical analysis, big O describes the error behavior of approximations.
 - ▶ Example: The Taylor expansion $\sin(x) = x - \frac{x^3}{6} + O(x^5)$ shows the error is bounded by a term proportional to x^5 for small x .

Understanding of Big O

- ▶ **Growth Comparison:** Big O measures how the magnitude of $f(x)$ compares to $g(x)$ as $x \rightarrow \infty$.
 - ▶ **Upper Bound:** It represents an asymptotic upper bound on the growth rate of $f(x)$, ignoring constant factors and lower-order terms.
1. Suppose $f(x) = 3x^2 + 5x + 7$. The dominant term as $x \rightarrow \infty$ is $3x^2$. Therefore, $f(x)$ is $O(x^2)$, because for large x , $|f(x)| \leq C \cdot |x^2|$ with $C \geq 3$.
 2. For $f(x) = \sin(x)$, $f(x)$ is $O(1)$, because $|\sin(x)| \leq 1$ for all x .

Properties

- ▶ **Additivity:** If $f(x) = O(g(x))$ and $h(x) = O(g(x))$, then $f(x) + h(x) = O(g(x))$.
- ▶ **Multiplication by Constants:** If $f(x) = O(g(x))$, then $c \cdot f(x) = O(g(x))$ for any constant c .
- ▶ **Transitivity:** If $f(x) = O(g(x))$ and $g(x) = O(h(x))$, then $f(x) = O(h(x))$.

Order Of Convergence

Definition (**Order of convergence**)

Let $\{p_n\}$ be a sequence that converges to a number p . Let $e_n = p_n - p$ for $n \geq 0$. If there exists positive constant λ and α such that

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|^\alpha} = \lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^\alpha} = \lambda$$

then $\{p_n\}$ is said to converge to p of **order** ' α ' with asymptotic error constant ' λ '

Remark 1: A sequence that converges of order ' α ', the error satisfies the asymptotic relation $|e_{n+1}| \approx \lambda |e_n|^\alpha$.

Remark 2:

- ▶ $\alpha = 1$: **Linear Convergence**
- ▶ $\alpha = 2$: **Quadratic Convergence**
- ▶ $\alpha = 3$: **Cubic Convergence**

Comparison between Linear, Quadratic, and Cubic order of convergence

Let each method has an asymptotic error constant as $\lambda = 0.5$ and error in initial approximation $e_0 = 1$

Approx	Linear	Quadratic	Cubic
	$(e_{n+1} \approx 0.5 e_n)$	$(e_{n+1} \approx 0.5 e_n ^2)$	$(e_{n+1} \approx 0.5 e_n ^3)$
$ e_1 $	0.5	0.5	0.5
$ e_2 $	0.25	0.125	0.0625
$ e_3 $	0.125	7.185×10^{-3}	1.2207×10^{-4}
$ e_4 $	0.0625	3.0518×10^{-5}	9.0949×10^{-13}
$ e_5 $	0.03125	4.6566×10^{-10}	3.7616×10^{-37}

Table 1: Error approximations for different convergence rates

asymptotic error constant

Definition

The asymptotic error constant, denoted C , appears in the error expression for iterative methods when analyzing the behavior of errors in consecutive iterations. For an iterative sequence $\{x_k\}$ converging to a solution x^* , the error $e_k = |x_k - x^*|$ satisfies:

$$e_{k+1} \approx C \cdot e_k^p,$$

where:

- ▶ $e_k = |x_k - x^*|$ is the error at the k -th iteration,
- ▶ p is the **order of convergence**,
- ▶ C is the **asymptotic error constant**.

Interpretation

- ▶ **Error Scaling:** The constant C quantifies how much the error is scaled during each iteration. Smaller values of C imply faster error reduction.
- ▶ **Relationship to p :** While p indicates the overall speed of convergence (e.g., linear, quadratic), C gives the proportionality factor governing the actual size of the errors.

1. Linear Convergence ($p = 1$):

$$e_{k+1} \approx C \cdot e_k.$$

Here, C directly determines how quickly the error reduces. For example, if $C = 0.5$, the error halves with each iteration.

2. Quadratic Convergence ($p = 2$):

$$e_{k+1} \approx C \cdot e_k^2.$$

The error decreases more rapidly, but C influences the speed of this quadratic reduction.

How it helps

- ▶ **Algorithm Efficiency:** Understanding C helps compare iterative methods with the same order of convergence. A method with a smaller C will perform better in practice.
- ▶ **Stopping Criteria:** Knowing C aids in estimating the number of iterations required to achieve a desired error tolerance.

Determine The Order Of Convergence

Recursive scheme

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right)$$

is used to compute an approximation to the square root of positive real number a .

Order of convergence = ?

Error at the $(n+1)^{th}$ term $= x_{n+1} - \sqrt{a}$

Error at the n^{th} term $= x_n - \sqrt{a}$

$$x_{n+1} - \sqrt{a} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right) - \sqrt{a}$$

$$x_{n+1} - \sqrt{a} = \frac{x_n^2 - 2x_n\sqrt{a} + a}{2x_n}$$

Determine The Order Of Convergence

$$x_{n+1} - \sqrt{a} = \frac{(x_n - \sqrt{a})^2}{2x_n}$$

$$\frac{x_{n+1} - \sqrt{a}}{(x_n - \sqrt{a})^2} = \frac{1}{2x_n}$$

taking $\lim_{n \rightarrow \infty}$ on both sides.

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1} - \sqrt{a}|}{|x_n - \sqrt{a}|^2} = \lim_{n \rightarrow \infty} \frac{1}{2x_n} = \frac{1}{2\sqrt{a}} \quad (\text{since } x_n \rightarrow \sqrt{a})$$

Order of convergence = 2

Asymptotic error constant = $\frac{1}{2\sqrt{a}}$

Exercise:

The sequence $x_{n+1} = \frac{1}{2} \left(x_n + \frac{9}{x_n} \right)$ converges to 3 (Take $a = 9$).

Compute e_n and $\frac{e_n}{|e_{n-1}|^2}$ for $n = 1, 2, 3, 4, 5$ and 6.

Rate of Convergence

Definition

Let the sequence $\{x_n\}$ converges to a number L . If there exists a sequence $\{\beta_n\}$ that converges to zero and a positive constant λ , independent of n , such that

$$|x_n - L| \leq \lambda |\beta_n|$$

for all sufficiently large value of n , then $\{x_n\}$ is said to converge to L with **Rate of Convergence** $O(\beta_n)$.

The sequence $\{\beta_n\}$ is taken to be of the form $1/n^a$ or $1/a^n$ for some positive constant a .

Comparing Rates of Convergence

Consider the sequences, $a_n = \frac{n+3}{n+7}$ and $b_n = \frac{2^n+3}{2^n+7}$

n	a_n	b_n
1	0.5000000000	0.5555555556
2	0.5555555556	0.6363636364
3	0.6000000000	0.7333333333
4	0.6363636364	0.8260869565
5	0.6666666667	0.6666666667
6	0.6923076923	0.9436619718
7	0.7142857143	0.9703703704
8	0.7333333333	0.9847908745
9	0.7500000000	0.9922928709

Table 2: Corresponding Terms in Two Sequences that Converge to 1

Comparing Rates of Convergence

$$a_n = \frac{n+3}{n+7}$$

$$\text{so } \lim_{n \rightarrow \infty} a_n = 1$$

$$\text{now, } |a_n - 1| = \left| \frac{n+3}{n+7} - 1 \right| = \frac{4}{n+7} \leq 4 \cdot \frac{1}{n}$$

Hence, we take $\lambda = 4$ and $\beta = \frac{1}{n}$ in the definition of rate of convergence. It follows that the sequence $\{a_n\}$ converges to 1 with **rate of convergence** $O(\frac{1}{n})$

Comparing Rates of Convergence

$$b_n = \frac{2^n + 3}{2^n + 7}$$

calculate the rate of convergence and λ .

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calculate the rate of convergence and λ .

$$\text{so } \lim_{n \rightarrow \infty} b_n = 1$$

$$\text{now, } |b_n - 1| = \left| \frac{2^n + 3}{2^n + 7} - 1 \right| = \frac{4}{2^n + 7} \leq 4 \cdot \frac{1}{2^n}$$

Comparing Rates of Convergence

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calculate the rate of convergence and λ .

$$\text{so } \lim_{n \rightarrow \infty} b_n = 1$$

$$\text{now, } |b_n - 1| = \left| \frac{2^n + 3}{2^n + 7} - 1 \right| = \frac{4}{2^n + 7} \leq 4 \cdot \frac{1}{2^n}$$

Hence, we take $\lambda = 4$ and $\beta = \frac{1}{2^n}$ in the definition of rate of convergence. It follows that sequence $\{b_n\}$ converges to 1 with **rate of convergence** $O(\frac{1}{2^n})$

Rate of Convergence of a Function

Definition

Let f be a function defined on $[a, b] \ni x = 0$ and suppose that $\lim_{x \rightarrow 0} f(x) = L$. If there exists a function g for which $\lim_{x \rightarrow 0} g(x) = 0$ and a positive constant k such that

$$|f(x) - L| \leq k|g(x)|$$

for sufficiently large value of x , then $f(x)$ is said to converge to L with **rate of convergence** $O(g(x))$.

Example

Consider the function $f(x) = \frac{\cos x - 1 + \frac{1}{2}x^2}{x^4}$. Find the $\lim_{x \rightarrow 0} f(x)$ and rate of convergence of f .

solution

We know that

$$\cos x = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \frac{1}{720}x^6 \cos \zeta$$

for some ζ between 0 and x . Hence,

$$\frac{\cos x - 1 + \frac{1}{2}x^2}{x^4} = \frac{1}{24} - \frac{1}{720}x^2 \cos \zeta.$$

Finally,

$$\left| \frac{\cos x - 1 + \frac{1}{2}x^2}{x^4} - \frac{1}{24} \right| = \frac{1}{720} |x^2 \cos \zeta| \leq \frac{1}{720} |x^2|,$$

Hence, $\lim_{n \rightarrow \infty} f(x) = \frac{1}{24}$, we take $k = \frac{1}{720}$ and $g(x) = x^2$ in the definition of rate of convergence. It follows that rate of convergence of $f(x)$ is $O(x^2)$. □